Boss Challenge — Paper 42 (Class 7)

Winds, Storms & Cyclones | Electric Current & its Effects | Algebraic Expressions | Data Handling

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

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1. The push that the air all around us exerts on every unit of area of a surface is called:

- **(A)** air pressure
- **(B)** wind speed
- **(C)** air temperature
- **(D)** humidity

2. A long, insulated wire wound into many turns that behaves like a magnet only while current flows is an:

- **(A)** electromagnet
- **(B)** compass
- **(C)** permanent magnet
- **(D)** battery

3. A letter such as x or n that can stand for different number values is called a:

- **(A)** constant
- **(B)** answer
- **(C)** variable
- **(D)** equation

4. Information collected together in the form of numbers or facts, ready to be studied, is called:

- **(A)** an average
- **(B)** data
- **(C)** a graph
- **(D)** a ruler

5. Wind is simply moving air, and it always flows from a region of high air pressure towards a region of:

- **(A)** higher ground
- **(B)** high air pressure
- **(C)** low air pressure
- **(D)** equal air pressure

6. When an electric current flows through a wire, one thing that always happens to the wire is that it:

- **(A)** becomes invisible
- **(B)** turns into water
- **(C)** becomes hot
- **(D)** becomes longer and longer

7. A combination of variables and constants joined by operations such as + and x, like 3x + 5, is an:

- **(A)** bar graph
- **(B)** plain sentence
- **(C)** algebraic expression
- **(D)** single constant

8. The number you get by adding up all the values in a set and dividing by how many there are is the:

- **(A)** mode
- **(B)** mean (average)
- **(C)** range
- **(D)** median

9. When air is warmed it expands, becomes lighter than the cooler air around it, and so it tends to:

- **(A)** turn into water
- **(B)** rise upward
- **(C)** sink downward
- **(D)** stay perfectly still

10. A short, thin piece of wire that melts and breaks the circuit when too much current flows is called a:

- **(A)** filament
- **(B)** fuse
- **(C)** battery
- **(D)** switch

11. In the term 5x, the number 5 multiplying the variable is called the:

- **(A)** variable
- **(B)** coefficient
- **(C)** constant term
- **(D)** exponent

12. In a set of data, the value that occurs the greatest number of times is called the:

- **(A)** range
- **(B)** mode
- **(C)** mean
- **(D)** median

13. Cooler, heavier air rushing in to take the place of warm air that has risen is exactly what we feel as a:

- **(A)** sunlight
- **(B)** rain
- **(C)** a shadow
- **(D)** wind

14. In an electric circuit, current keeps flowing only while the path it travels along is:

- **(A)** very short
- **(B)** open (broken)
- **(C)** closed (complete)
- **(D)** painted blue

15. Terms that contain exactly the same variables raised to the same powers, such as $3x$ and $5x$, are called:

- **(A)** coefficients
- **(B)** like terms
- **(C)** unlike terms
- **(D)** constants

16. When a set of values is arranged in order, the value lying right in the middle is called the:

- **(A)** median
- **(B)** range
- **(C)** mean
- **(D)** mode

17. An experiment shows that the faster air moves over a surface, the air pressure there becomes:

- **(A)** exactly the same
- **(B)** higher
- **(C)** lower
- **(D)** zero

18. Within a glowing electric bulb, the slender coiled wire that lights up as current passes through is the:

- **(A)** filament
- **(B)** socket
- **(C)** fuse
- **(D)** switch

19. In the algebraic expression $3x + 5$, the part that has no variable, the plain number, is the:

- **(A)** exponent
- **(B)** coefficient
- **(C)** constant term
- **(D)** variable term

20. The mean of the three numbers 4, 6 and 8 is:

- **(A)** 4
- **(B)** 18
- **(C)** 9
- **(D)** 6

21. Blow hard through the gap between two balloons hanging side by side and, surprisingly, the balloons swing:

- **(A)** not move at all
- **(B)** straight upward
- **(C)** towards each other
- **(D)** apart from each other

22. Passing an electric current through water can break it up and is shown by the slow rise of:

- **(A)** flames of fire
- **(B)** bubbles of gas
- **(C)** grains of salt
- **(D)** blocks of ice

23. The algebraic expression that means 'seven more than a number x' is written as:

- **(A)** $x - 7$
- **(B)** $x + 7$
- **(C)** $7/x$
- **(D)** $7x$

24. Wind speeds on five days were 20, 30, 40, 50 and 60 km/h. Their mean (average) wind speed was:

- **(A)** 200 km/h
- **(B)** 40 km/h
- **(C)** 50 km/h
- **(D)** 60 km/h

25. The single biggest reason winds blow across the whole Earth is the uneven:

- **(A)** weight of the clouds
- **(B)** spinning of windmills
- **(C)** colour of the oceans
- **(D)** heating of the Earth by the Sun

26. The repeated striking of the hammer on the gong in an electric bell happens because its electromagnet keeps getting:

- **(A)** magnetised and demagnetised
- **(B)** hotter and hotter
- **(C)** bigger and bigger
- **(D)** heavier and heavier

27. Simplifying the expression $2x + 3x$ by adding the like terms gives:

- **(A)** $5x$
- **(B)** $5x^2$
- **(C)** $6x$
- **(D)** $23x$

28. A bar graph shows and compares data using the heights or lengths of its:

(A) dots
(B) colours only
(C) titles
(D) bars

29. By day the land heats up faster than the sea, so cool air blows in from the sea towards the land as a:

(A) land breeze
(B) cyclone
(C) monsoon
(D) sea breeze

30. The larger the electric current you send through a given wire, the more the wire produces of:

(A) heat
(B) sound
(C) light only and no heat
(D) ice

31. The result of the subtraction $7x - 4x$ is:

(A) $28x$
(B) $11x$
(C) $3x$
(D) 3

32. The difference between the highest value and the lowest value in a set of data is called the:

(A) range
(B) mean
(C) mode
(D) median

33. At night the land cools faster than the sea, so the breeze reverses and now blows from the land towards:

(A) the hills
(B) straight upward
(C) the sea
(D) the land

34. A modern safety device used in homes in place of a fuse, which can simply be switched back on, is the:

(A) compass
(B) filament
(C) anemometer
(D) MCB (miniature circuit breaker)

35. An algebraic expression that has exactly one term, such as 5x or 7, is called a:

(A) binomial
(B) equation
(C) monomial
(D) trinomial

36. During a cyclone the strongest gust was 120 km/h and the weakest 40 km/h. The range of the gust speeds was:

(A) 40 km/h
(B) 160 km/h
(C) 120 km/h
(D) 80 km/h

37. The seasonal winds that sweep in from the ocean and bring heavy rains to much of India are the:

(A) tornado winds
(B) trade winds
(C) land breezes
(D) monsoon winds

38. Several electric cells joined together end to end to give a larger supply form a:

(A) single switch
(B) conductor
(C) battery
(D) fuse

39. An algebraic expression made up of exactly two terms, such as 3x + 5, is called a:

(A) monomial
(B) binomial
(C) trinomial
(D) variable

40. The mode of the data set 2, 3, 3, 5 and 7 is:

(A) 3
(B) 7
(C) 5
(D) 2

41. A very violent storm in which high-speed winds spiral round and round a low-pressure centre is called a:

(A) sea breeze
(B) rainbow
(C) cyclone
(D) monsoon

42. An electromagnet is special because it stops being a magnet the moment the current is:

(A) made longer
(B) warmed up
(C) painted red
(D) switched off

43. An algebraic expression that contains exactly three terms, such as $x + y + 5$, is called a:

(A) monomial
(B) constant
(C) binomial
(D) trinomial

44. The mean of the first five counting numbers 1, 2, 3, 4 and 5 is:

(A) 15
(B) 5
(C) 3
(D) 2.5

45. At the very middle of a cyclone there is a strangely calm, low-pressure region known as the:

(A) tail
(B) crest
(C) eye
(D) root

46. Copper and the other metals that readily allow an electric current to flow through them are known as:

(A) fuses
(B) insulators
(C) conductors
(D) magnets

47. The value of the expression $2x + 1$ when $x = 3$ is:

(A) 7
(B) 5
(C) 9
(D) 6

48. A home used 5, 5, 6 and 8 units of electricity on four days. The mean daily use was:

(A) 24 units
(B) 8 units
(C) 6 units
(D) 5 units

49. A storm exactly like a cyclone, when it forms over the seas near America, is instead given the name:

- **(A)** tsunami
- **(B)** drought
- **(C)** blizzard
- **(D)** hurricane

50. A device that changes stored chemical energy into the electrical energy that drives a current is a:

- **(A)** wire
- **(B)** bulb
- **(C)** switch
- **(D)** cell

51. The value of the expression 5a when a = 4 is:

- **(A)** 20
- **(B)** 1
- **(C)** 9
- **(D)** 54

52. A number that measures how likely an event is to happen, always between 0 and 1, is called its:

- **(A)** range
- **(B)** mode
- **(C)** probability
- **(D)** mean

53. A dark, funnel-shaped column of very fast spinning air that reaches down from a storm cloud to the ground is a:

- **(A)** rainbow
- **(B)** glacier
- **(C)** sea breeze
- **(D)** tornado

54. A current-carrying coil can pick up steel pins and then drop them when switched off; this shows that current has a:

- **(A)** magnetic effect
- **(B)** freezing effect
- **(C)** cooling effect
- **(D)** sound effect

55. The algebraic expression for 'twice a number n, decreased by 5' is:

- **(A)** n - 5
- **(B)** 5 - 2n
- **(C)** 2n + 5
- **(D)** 2n - 5

56. When a fair coin is tossed once, the probability of getting heads is:

- **(A)** 0
- **(B)** 2
- **(C)** 1
- **(D)** 1/2

57. Cyclones get their huge energy and usually begin to form over:

- **(A)** dry deserts
- **(B)** cold mountain peaks
- **(C)** warm ocean water
- **(D)** frozen polar ice

58. The wire of an electric heater glows red-hot when switched on; this is a clear example of the:

- **(A)** heating effect of current
- **(B)** cooling effect of current
- **(C)** magnetic effect of current
- **(D)** chemical effect of current

59. Adding the two expressions $3a + 2$ and $5a + 4$ gives:

- **(A)** $15a + 6$
- **(B)** $8a + 6$
- **(C)** $8a + 8$
- **(D)** $8a^2 + 6$

60. A graph that draws two bars side by side for each item is especially useful for:

- **(A)** hiding the data
- **(B)** drawing circles
- **(C)** comparing two sets of data
- **(D)** measuring time only

61. The instrument made to measure the speed of the wind is called the:

- **(A)** barometer
- **(B)** anemometer
- **(C)** rain gauge
- **(D)** thermometer

62. A bulb draws a current of i amperes. The total current drawn by 3 such identical bulbs, all the same, is:

- **(A)** i amperes
- **(B)** i + 3 amperes
- **(C)** 3i amperes
- **(D)** i/3 amperes

63. Subtracting 2x from 7x leaves:

- **(A)** 14x
- **(B)** 5x
- **(C)** 5
- **(D)** 9x

64. Arranged in order, the median of the five values 3, 5, 7, 9 and 11 is:

- **(A)** 9
- **(B)** 5
- **(C)** 7
- **(D)** 11

65. Just around the calm eye of a cyclone lies the ring of the fiercest winds and heaviest rain, called the:

- **(A)** eye wall
- **(B)** eye
- **(C)** base
- **(D)** tail

66. The filament of a bulb is made from a metal chosen because it has a very:

- **(A)** low melting point
- **(B)** bright colour when cold
- **(C)** pleasant smell
- **(D)** high melting point

67. A storm moves d kilometres each hour at a steady pace. In 5 hours it moves a distance of:

- **(A)** d/5 kilometres
- **(B)** d kilometres
- **(C)** d + 5 kilometres
- **(D)** 5d kilometres

68. Records show that a town was hit by a cyclone in 2 of the last 10 years. The probability of a cyclone in a year is about:

- **(A)** 2/1
- **(B)** 1/5
- **(C)** 1/2
- **(D)** 10/2

69. The two conditions that join together to create a cyclone are high wind speeds and a difference in:

- **(A)** soil colour
- **(B)** air pressure
- **(C)** number of islands
- **(D)** ocean saltiness

70. To do its job of protecting a circuit, a fuse must be connected so that the whole circuit's current:

(A) only touches it slightly
(B) avoids it completely
(C) passes through it
(D) flows around it

71. In the term 4xy, the variables present are:

(A) x and y
(B) 4 and x and y
(C) only 4
(D) only x

72. In a set of data, the number of times a particular value appears is called its:

(A) frequency
(B) median
(C) range
(D) mean

73. A sudden, dangerous rise of seawater that floods low coastal land during a cyclone is called a:

(A) land breeze
(B) storm surge
(C) snow drift
(D) dew fall

74. The two effects of electric current you study at this level are the heating effect and the:

(A) magnetic effect
(B) smell effect
(C) sound effect
(D) freezing effect

75. The coefficient of the variable in the term -x is:

(A) 1
(B) x
(C) 0
(D) -1

76. For the even-sized set 2, 4, 6 and 8, the median is the mean of the two middle values, which is:

(A) 4
(B) 5
(C) 20
(D) 6

77. The safest action to take the moment an official cyclone warning is announced for your area is to:

- **(A)** move to a safer, stronger shelter
- **(B)** ignore the warning
- **(C)** switch on all taps
- **(D)** go out to watch the sea

78. Adding a soft-iron piece, the core, inside a current-carrying coil makes the electromagnet:

- **(A)** permanent forever
- **(B)** colder
- **(C)** weaker
- **(D)** stronger

79. In the expression 3x + 4y + 5x, the like terms that can be combined are:

- **(A)** 3x and 5x
- **(B)** all three terms
- **(C)** 4y and 5x
- **(D)** 3x and 4y

80. A fan ran for 4, 4, 5 and 6 hours on four days. The mode of these running hours is:

- **(A)** 4 hours
- **(B)** 6 hours
- **(C)** 19 hours
- **(D)** 5 hours

81. The brilliant flash of light seen across the sky during a thunderstorm is called:

- **(A)** thunder
- **(B)** lightning
- **(C)** a rainbow
- **(D)** a sunbeam

82. The chief job of an electric fuse in a household circuit is to protect the wiring and appliances from:

- **(A)** bright sunlight
- **(B)** loud noise
- **(C)** too much current
- **(D)** too little current

83. Written in the shortest algebraic form, 'the product of a and b' is:

- **(A)** ab
- **(B)** a/b
- **(C)** a - b
- **(D)** a + b

84. The mean of the four numbers 10, 20, 30 and 40 is:

(A) 25
(B) 100
(C) 30
(D) 40

85. The loud rumbling or crashing sound that reaches us a little after the lightning flash is the:

(A) an echo
(B) thunder
(C) lightning
(D) a whistle

86. A heater is used for the same number of hours, t hours, on each of 7 days. The total time it runs is:

(A) t hours
(B) t + 7 hours
(C) 7/t hours
(D) 7t hours

87. The value of x^2 (x squared) when $x = 5$ is:

(A) 52
(B) 25
(C) 7
(D) 10

88. The probability of an event that is absolutely certain to happen is:

(A) 100
(B) 1/2
(C) 1
(D) 0

89. Towering, dark thunderclouds build up mainly when warm, moist air over a heated area rapidly:

(A) rises upward
(B) sinks to the ground
(C) stops moving
(D) freezes solid

90. An ordinary magnetic compass needle, brought near a wire carrying a current, is seen to:

(A) catch fire
(B) melt at once
(C) stay perfectly still
(D) deflect (turn aside)

91. Three numbers have a mean (average) of m. Written algebraically, their total sum is:

- **(A)** m
- **(B)** m + 3
- **(C)** m/3
- **(D)** 3m

92. The probability of an event that can never possibly happen is:

- **(A)** -1
- **(B)** 0
- **(C)** 1
- **(D)** 1/2

93. A cyclone's wind speed rose from 90 km/h to 180 km/h as it strengthened. The new speed is how many times the old speed?

- **(A)** 2 times
- **(B)** the same
- **(C)** half
- **(D)** 3 times

94. Rubber and plastic are used to cover electric wires because such materials are good:

- **(A)** insulators
- **(B)** fuses
- **(C)** magnets
- **(D)** conductors

95. The expression for 'a number y multiplied by itself' is:

- **(A)** 2y
- **(B)** y/2
- **(C)** y + 2
- **(D)** y^2

96. Rain fell on three days, totalling 90 mm. The mean (average) daily rainfall over those days was:

- **(A)** 270 mm
- **(B)** 30 mm
- **(C)** 3 mm
- **(D)** 90 mm

97. Today, forming cyclones far out at sea are spotted and tracked from above by:

- **(A)** microscopes
- **(B)** satellites
- **(C)** fishing nets
- **(D)** telescopes pointed at stars

98. For a bulb in a torch to light up, the cell, the switch, the bulb and the wires must together form one:

- **(A)** pile of magnets
- **(B)** broken circuit
- **(C)** single wire alone
- **(D)** complete circuit

99. Removing the brackets, the expression 2(x + 3) is the same as:

- **(A)** 2x + 5
- **(B)** 2x + 6
- **(C)** x + 6
- **(D)** 2x + 3

100. When every observation is added up and the total is divided by the number of observations, the result is the:

- **(A)** mode
- **(B)** median
- **(C)** range
- **(D)** mean (arithmetic average)